



Section 1: Research Design

Reminder

Your teacher will mark Research Design on the clarity of your question, how well your variables are explained, and whether your method is detailed enough for another student to replicate without asking you anything.

⚠ IMPORTANT: READ BEFORE SUBMITTING ⚠

This is a working template. Before you submit your IA to your teacher:

1. **DELETE** all blue hint text and grey placeholder text in input boxes
2. **DELETE** the BioMate header, footer, and cover page before submission
3. **REPLACE** all placeholder text in square brackets [like this] with your actual data
4. **CHECK** that no tables cross a page break in your final document

The IB does not want to see this template. They want to see your completed investigation.

1.1 Research Question

💡 Your RQ must name the IV, the DV, and the organism or system. It must be answerable with numbers.

Research Question

e.g. How does [IV with range and units] affect [DV with measurement method] in [organism / system]?

1.2 Hypotheses

💡 H_0 always says: no significant effect. H_1 predicts a direction AND gives a biological reason.

H_0 — Null Hypothesis

There is no statistically significant effect of [IV] on [DV]...

H_1 — Alternate Hypothesis

There is a statistically significant [increase / decrease] in [DV] as [IV] increases / decreases, because [biological mechanism]...



1.3 Variables

Do not just name your variables. Explain HOW you will measure or control each one, with units.

Variable Type	Name it	Units + how measured / controlled
Independent Variable (IV)	Name the variable...	e.g. 100–500 lux, measured with lux meter...
Dependent Variable (DV)	Name the variable...	e.g. O ₂ bubbles per minute, counted by eye...

#	Controlled Variable	How I will control it (value + method + instrument)
CV 1	Name the controlled variable...	How will you control it? (specific value, method, instrument)...
CV 2	Name the controlled variable...	How will you control it? (specific value, method, instrument)...
CV 3	Name the controlled variable...	How will you control it? (specific value, method, instrument)...
CV 4	Name the controlled variable...	How will you control it? (specific value, method, instrument)...

1.4 Materials

List everything you need with quantities and specifications. If you use a 250 ml beaker, say 250 ml.

Materials List
List each item on a new line. Include quantities, concentrations, sizes, and model numbers where relevant...

1.5 Method

Number every step. Write it so a stranger could follow it exactly. Justify key choices (why this range? why n=5?).

Step-by-step Method
Step 1. ...Step 2. ...Step 3. ...(Add as many steps as you need. Justify each key decision in brackets.)

1.6 Safety and Ethics

Include this even for low-risk investigations. Mention COSHH data for chemicals and ethical treatment of organisms.